
JAL1501 MACHINE LEARNING

UNIT – I INTRODUCTION TO MACHINE LEARNING

UNIT-1 INTRODUCTION TO MACHINE LEARNING SYLLABUS

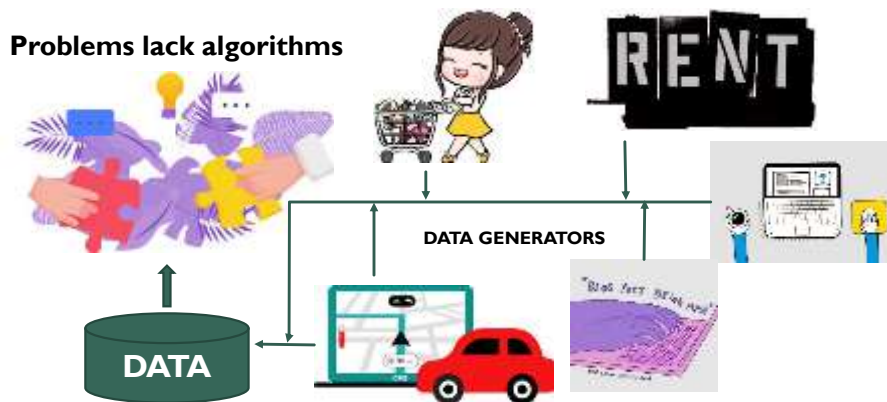
- Review of Linear Algebra for machine learning -
- Introduction and motivation for machine learning -
- Examples of machine learning applications - Vapnik -
- Chervonenkis (VC) dimension - Probably Approximately
- Correct (PAC) learning - Hypothesis spaces - Inductive bias -
- Generalization - Bias variance trade-off.

INTRODUCTION FOR MACHINE LEARNING

MACHINE LEARNING

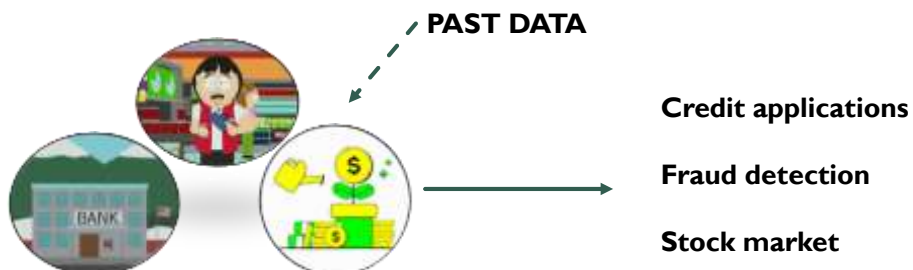
- ❑ Algorithms devised for many tasks
- ❑ Digital technology → has become the main means to store, process, and transmit information.
- ❑ Some tasks as humans do have no algorithm
- ❑ Eg. face recognition, game playing, conversations
- ❑ Machine learning → learning to do these types of things.
- ❑ Training data, testing data
- ❑ Start from a very general model with many parameters
- ❑ General model → can do all sorts of tasks depending on how its **parameters** are set.
- ❑ Learning → **adjusting** the values of those parameters so that the **model matches best with the data** it sees during **training**.
- ❑ Thus algorithm is obtained.

MACHINE LEARNING



MACHINE LEARNING

- Data mining → Application of machine learning methods to large databases.
- Large volume of earth and raw material – small amount of precious material;
- Large volume of data is processed → simple valuable model → eg. high predictive accuracy
- Application areas:



MACHINE LEARNING

Retail

Customer behavior changes in time and by geographic location – product purchase pattern analyzed



Medicine

Medical diagnosis



Telecommunication

Call patterns analyzed for network optimization and to maximize the QoS



MACHINE LEARNING

- Construction of a good and useful approximation
- Machine learning - helps us find solutions to many problems in vision, speech recognition, and robotics.
- Machine learning → programming computers to optimize a performance criterion using example data or past experience - uses the theory of statistics in building mathematical models. Then role of computer science??
- The model may be predictive or descriptive or both.

Role of computer science in machine learning

1. Training → Efficient algorithms needed to solve the optimization problem and to store and process the massive amount of data we generally have.
2. After learning the model, its representation and efficient algorithmic solution for inference needed - mainly space and time complexity - important as its predictive accuracy.



EXAMPLES OF MACHINE LEARNING APPLICATIONS



EXAMPLES OF MACHINE LEARNING APPLICATIONS

- ☐ Association Rules
- ☐ Classification
- ☐ Regression
- ☐ Unsupervised Learning
- ☐ Reinforcement Learning

EXAMPLES OF MACHINE LEARNING APPLICATIONS

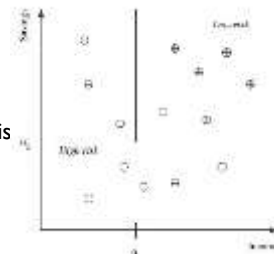
ASSOCIATION RULES

- ❑ In the case of **retail**
- ❑ Basket analysis (or) market-basket analysis → finding associations between products bought by customers
- ❑ If people who buy X (bread) typically also buy Y (jam), and if there is a customer who buys X and does not buy Y, they are a potential (possible) Y customer.
- ❑ To find association rules, we must find the conditional probability of the form $P(Y|X)$ → the probability that customers buying X also buy Y
- ❑ We can also find $P(Y|X, D)$ - D → set of customer attributes, for example, gender, age, and marital status

EXAMPLES OF MACHINE LEARNING APPLICATIONS

CLASSIFICATION

- ❑ In the case of **credit scoring**
 - ❑ Risk in providing loan / credit
 - ❑ Uses the customer's financial capacity, such as income, savings, collaterals, profession, age, and past financial his
 - ❑ Customers classified as low-risk and high-risk customers
 - ❑ Classification rule obtained: it fits the past data
- IF income > 01 AND savings > 02 THEN low-risk ELSE high-risk*
- If the future is similar to the past, then we can make correct predictions for novel instances.
 - It is decided whether to accept or refuse the loan depending on the possible gain and loss



EXAMPLES OF MACHINE LEARNING APPLICATIONS

CLASSIFICATION

- ❑ In the case of **pattern recognition**
 - ❑ Optical character recognition, which is recognizing character codes from their images.
 - ❑ Face recognition
 - ❑ Medical diagnosis
 - ❑ Speech recognition
 - ❑ Spam filtering
 - ❑ Sentiment prediction → happiness of customers
 - ❑ Biometrics → recognition or authentication of people using their physiological and/or behavioral characteristics
 - ❑ Outlier detection → eg, fraud detection or novelty detection

EXAMPLES OF MACHINE LEARNING APPLICATIONS

REGRESSION

- ❑ Supervised learning problems → Given an input, X , and an output, Y , learn the mapping from the input to the output.
- ❑ Eg. Predicting the price of a used car → with the car attributes as inputs (brand, year, engine capacity, mileage, etc. that we believe affect a car's worth.
- ❑ Survey on past transactions – collect the training dataset
- ❑ the machine learning program for regression fits a function to these data to learn Y as a function of X .
- ❑ Eg. $y = mx + m_0$
- ❑ Regression → used to identify the associations between variables occurring in some data → shows the magnitude of such an association and determine its statistical significance.

EXAMPLES OF MACHINE LEARNING APPLICATIONS

REGRESSION

- ❑ Regression → a statistical method used in finance, investing, and other disciplines that attempts to determine the strength and character of the relationship between a dependent variable and one or more independent variables.
- ❑ Coined in the 19th century to describe the statistical feature of biological data
- ❑ Regression model expressed as $y = g(x | \theta)$ where $g()$ → regression function of the model and θ → set of parameters
- ❑ The parameters are optimized - such that the approximation error is minimized (our estimates are as close as possible to the correct values given in the training set).
- ❑ An example quadratic model: $y = w_2x^2 + w_1x + w_0$

EXAMPLES OF MACHINE LEARNING APPLICATIONS

REGRESSION

- ❑ Eg. navigation of a mobile robot or an autonomous car –
 - ❑ Inputs?? → sensors, camera
 - ❑ Outputs?? → steering angle, speed, brake usage
 - ❑ Training data ?? → from human drivers' experience
- ❑ Eg. Recommendation system for movies – based on attributes like genre, actors, etc. ranking function is generated

EXAMPLES OF MACHINE LEARNING APPLICATIONS

UNSUPERVISED LEARNING

- ❑ Unsupervised learning → only input data available
- ❑ Aim → to find the regularities in the input → Identifying certain patterns that occur more often than others → termed as density estimation.
- ❑ Clustering → finding clusters or groupings of input
 - ❑ Eg. customer segmentation → finding the types of frequent customers in a company to enhance CRM (customer relationship management.)
 - ❑ Eg. Document clustering → group similar documents → documents (or bag of words)

EXAMPLES OF MACHINE LEARNING APPLICATIONS

REINFORCEMENT LEARNING

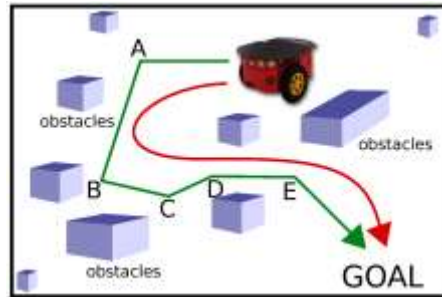
- ❑ Applications with a sequence of actions – where the goodness of a single action is more important than the action itself.
- ❑ Good action is a part of a good policy
- ❑ Machine learning → assessing the goodness of policies – learn from the past good action sequences
- ❑ Reinforcement learning (RL) is a machine learning (ML) technique that trains software to make decisions to achieve the most optimal results. It mimics the trial-and-error learning process that humans use to achieve their goals. Software actions that work towards your goal are reinforced, while actions that detract from the goal are ignored.

EXAMPLES OF MACHINE LEARNING APPLICATIONS

REINFORCEMENT LEARNING



Game playing



A robot navigating in an environment
in search of a goal location

MACHINE LEARNING – AN INTRODUCTON

MACHINE LEARNING

Definitions:

- ❑ ML is a subset of AI which combines data with statistical tools to predict an output which can be used to make actionable insights.
- ❑ Machine learning (ML) is a branch of artificial intelligence (AI) and computer science that focuses on using the data and algorithms to enable AI to imitate the way that humans learn, gradually improving its accuracy.
- ❑ Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and generalize to unseen data and thus perform tasks without explicit instructions

MACHINE LEARNING

Definitions:

- ❑ Machine learning is programming computers to optimize a performance criterion using example data or past experience. We have a model defined up to some parameters, and learning is the execution of a computer program to optimize the parameters of the model using the training data or past experience. The model may be predictive to make predictions in the future, or descriptive to gain knowledge from data, or both.
- ❑ ML is the field of study concerned with the question of how to construct computer programs that automatically improve with experience.

MACHINE LEARNING

Application areas of Machine Learning

- ❑ Augmentation
- ❑ Automation
- ❑ Health industry
- ❑ Marketing
- ❑ Supply chain

Importance of machine Learning

- ❑ ML can take **decision** with minimal human intervention.
- ❑ Machines make all these things possible by **filtering** useful pieces of information and piecing them together based on patterns to get accurate results.
- ❑ ML is important because it gives enterprises a view of trends in **customer behavior** and **operational business patterns**, as well as supports the development of **new products**.

MACHINE LEARNING

Popular Machine Learning Applications

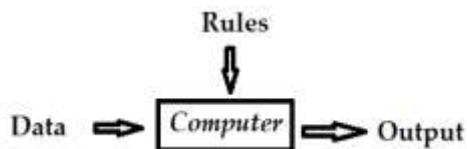
- | | |
|---|--|
| ■ Social Media Features | ■ Predict Potential Heart Failure |
| ■ Product Recommendations | ■ Banking and Finance Domain |
| ■ Image Recognition | ■ Language Translation |
| ■ Sentiment Analysis | ■ Transportation and Traffic Prediction |
| ■ Automating Employee Access Control | ■ Virtual Personal Assistance |
| ■ Marine Wildlife Preservation | ■ Email Spam and Malware detection & Filtering |
| ■ Regulating Healthcare Efficiency and Medical Services | ■ Credit card fraud detection |
| | ■ Stock Marketing and Trading |

MACHINE LEARNING

Traditional Programming Vs Machine Learning

Traditional Programming

- ❑ Programmer consults the company / domain expert and code the rules with logical foundation.
- ❑ For complex problems, new rules must be constructed by the programmer



Machine Learning

- ❑ Machine learns the i/p and o/p and generates the new rules and adapt in response to new data and experiences improved efficiency over time.



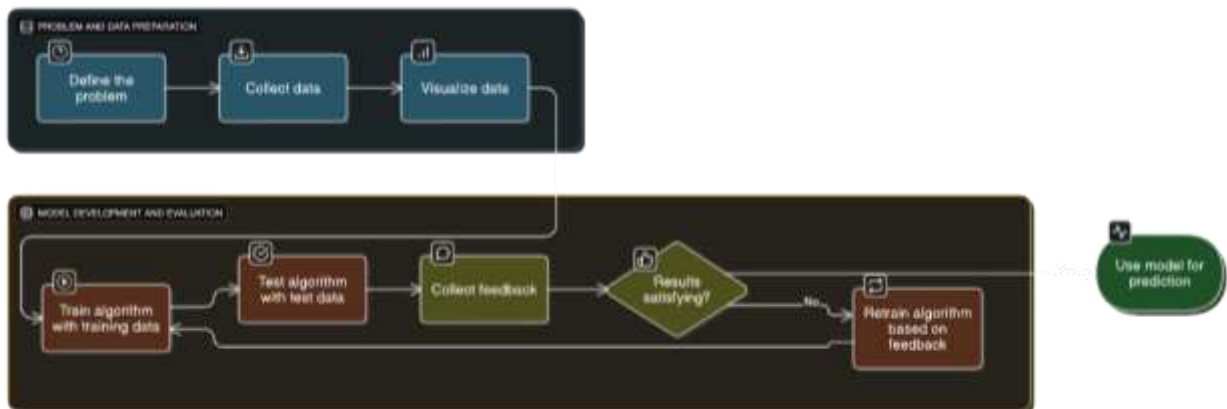
MACHINE LEARNING

How Does Machine Learning Work?

- ❑ The Machine Learning process starts with inputting training data (past data) into the selected algorithm. Training data being known or unknown data to develop the final ML algorithm. The type of training data input has an impact on the algorithm.
- ❑ **New input data** is fed into the machine learning algorithm to test whether the algorithm works correctly.
- ❑ The prediction and results are then **checked** against each other.
- ❑ If the prediction and results don't match, the algorithm is **re-trained** multiple times until the data scientist gets the desired outcome.
- ❑ This enables the machine learning algorithm to continually learn on its own and produce the optimal answer, gradually increasing in accuracy over time.

MACHINE LEARNING

MACHINE LEARNING PROCESS



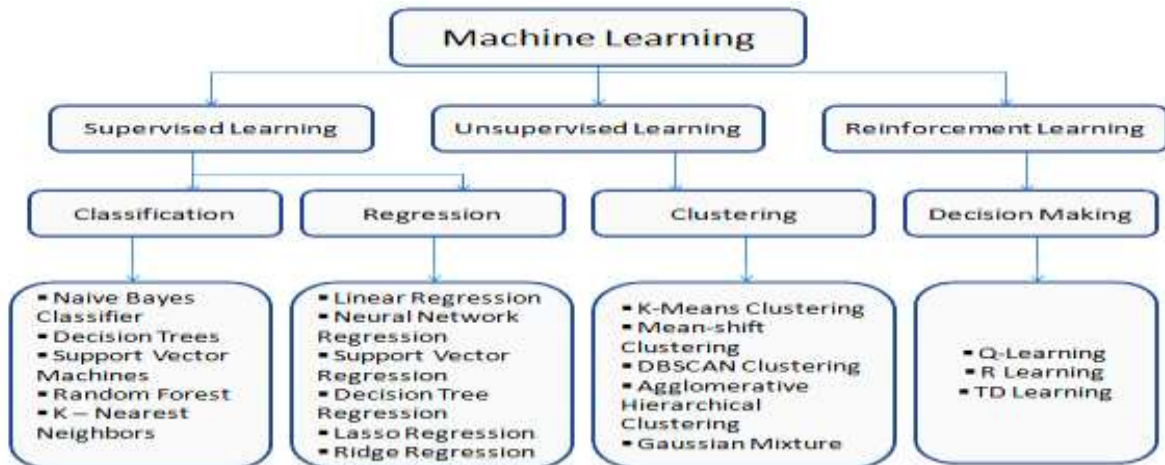
MACHINE LEARNING

When is Machine Learning used?

- ☐ When human expertise does not exist (navigating on a planet)
- ☐ When human can't explain themselves to the expertise (speech recognition)
- ☐ When models must be customized (personalized medicine, jobs)
- ☐ When models are based on massive amount of data (Genomics – study about DNA)

MACHINE LEARNING

Classification of Machine Learning



MACHINE LEARNING

I. Supervised Learning

- ❑ In supervised learning, **known or labeled data** is used for the **training** data.
- ❑ Since the data is known, the learning is, therefore, supervised, i.e., directed into successful execution.
- ❑ The input data goes through the ML algorithm and is used to train the model.
- ❑ Once the model is trained based on the known data, you can use unknown data into the model and get a new response.
- ❑ Eg. A model is trained with labelled dataset of images of animals, fruits, furniture, etc. when a new image comes as input, it is compared with labeled dataset and predicts the correct label.

MACHINE LEARNING

Classification

- ❑ In machine learning, classification is the problem of identifying to which of a set of categories a new observation belongs, on the basis of a training set of data containing observations (or instances) whose category membership is known.
- ❑ Algorithm can predict a discrete value that can identify the new input data as a member of a particular class or group.
- ❑ Eg. Classifying a dog image as a 'dog', Optical character recognition, Face recognition, Speech recognition, Medical diagnosis, Spam detection, biometric authentication, churn prediction, Predicting student performance, defect detection
- ❑ Based on a classification function
- ❑ Classification algorithms: a) Logistic regression b) Naive Bayes algorithm c) k-NN algorithm d) Decision tree algorithm e) Support vector machine algorithm f) Random forest algorithm

MACHINE LEARNING

Regression

- ❑ In machine learning, a regression problem is the problem of predicting the value of a numeric variable based on observed values of the variable.
- ❑ The value of the output variable may be a number, such as an integer or a floating point value.
- ❑ Data are often quantities, such as amounts and sizes.
- ❑ The input variables may be discrete or real-valued.
- ❑ Algorithm can predict continuous data for a new input.
- ❑ Eg. predicting the price of a land, or any other used product, weather forecasting
- ❑ Based on a regression function
- ❑ Types: a) Simple linear regression, b) Multivariate linear regression, c) Polynomial regression, d) Logistic regression

MACHINE LEARNING

2. Unsupervised Learning

- ❑ In unsupervised learning, the training data is unknown and unlabeled - meaning that no one has looked at the data before.
- ❑ Without the aspect of known data, the input cannot be guided to the algorithm, which is where the unsupervised term originates from.
- ❑ This data is fed to the Machine Learning algorithm and is used to train the model.
- ❑ The trained model tries to search for a pattern and give the desired response.
- ❑ It tries to put all inputs together so that you get the same things in similar groups.

MACHINE LEARNING

Clustering

- ❑ In machine learning, clustering is an unsupervised learning technique used to group similar data points into clusters without using labeled data..
- ❑ No predefined labels or categories.
- ❑ The algorithm discovers patterns in the data
- ❑ Goals:
 - ❑ To maximize intra-cluster similarity (within a cluster).
 - ❑ To minimize inter-cluster similarity (between different clusters).
- ❑ Visualised using scatter plots
- ❑ Eg. Customer / image Segmentation, Document or Text Clustering, Anomaly or Outlier Detection, Social Network Analysis, Recommender Systems
- ❑ Clustering algorithms: K-means, Hierarchical clustering, DBSCAN, Gaussian Mixture Models
- ❑ Applications: Customer segmentation, Document or text clustering, Image compression, Market basket analysis, Anomaly detection

MACHINE LEARNING

3. Reinforcement Learning

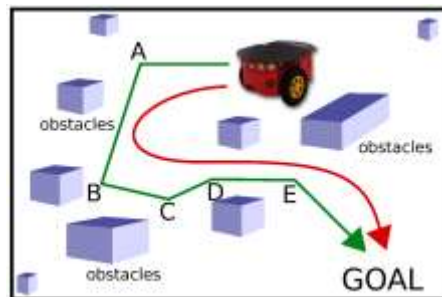
- ❑ A type of machine learning where an agent learns to make decisions by interacting with an environment, receiving rewards or penalties based on its actions.
- ❑ The agent observes a state, takes an action, and receives a reward.
- ❑ It updates its policy to maximize the cumulative reward over time
- ❑ Learn the optimal policy that maximizes the total reward in the long run.
- ❑ Eg. Robotics, Game playing, Recommendation (Personalized content suggestions), Finance, Smart Systems (Traffic signal control), Optimized treatment strategies.
- ❑ Algorithms: Value-Based Methods, Policy-Based Methods, Model-Based Methods, Deep Reinforcement Learning.

EXAMPLES OF MACHINE LEARNING APPLICATIONS

REINFORCEMENT LEARNING



Game playing



A robot navigating in an environment in search of a goal location

VAPNIK - CHERVONENKIS (VC) DIMENSION

VAPNIK - CHERVONENKIS (VC) DIMENSION

- ❑ Used to quantify how powerful is the model.
- ❑ **Definition** → VC Dimension is a concept in statistical learning theory that measures the capacity of a model to fit different datasets. It essentially quantifies how well a model can learn to classify data points in various ways.
- ❑ **VC(H) → The maximum number of points that can be shattered by H, where H is the hypothesis.** Eg. for H → separating +ve values from – values
- ❑ Introduced by Vladimir Vapnik and Alexey Chervonenkis.
- ❑ Helps in understanding the trade-off between model complexity and generalization ability.
- ❑ In designing real world applications, models tried one by one and the accuracy of each is evaluated – the one with highest accuracy is chosen
- ❑ But in statistical approach – VC dimension used to find the powerful model

VAPNIK - CHERVONENKIS (VC) DIMENSION

- A function is found for H and the capacity of H is measured.
- **VC dimension of a model** $\rightarrow$ maximum number of points that can be separated by a model (a straight line / a rectangle / a circle) for all possible configurations.

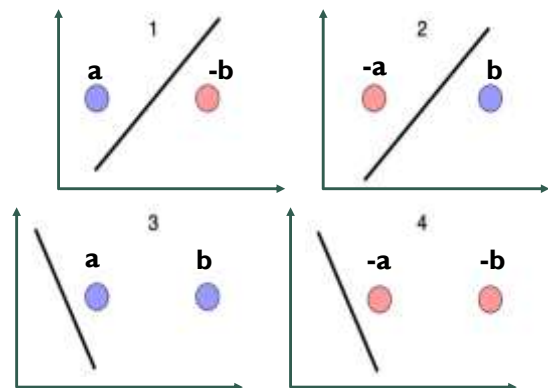
Hypotheses

- A hypothesis is a **function or model** that **approximates** the relationship between input data (features) and the target variable (output).
- It is essentially the candidate solution that the learning algorithm generates and refines during **training**.
- The hypothesis is **used** to make predictions on **new, unseen data**.

VAPNIK - CHERVONENKIS (VC) DIMENSION

Eg. 2-points Linear Classifier Model in a 2-D plane

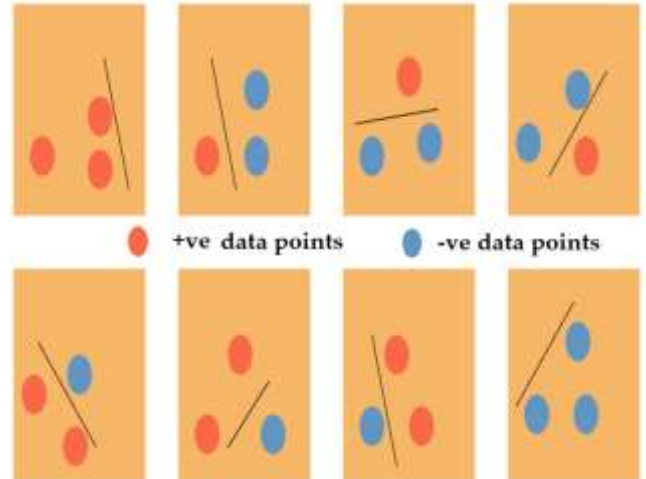
- No. of data points = $N = 2$ (+ve & -ve)
- No. of possibilities = $2^N = 2^2 = 4$
- If the points are a and b , the 4 possibilities are (a,b) , $(-a,b)$, $(a,-b)$, $(-a,-b)$
- A **straight line can separate** 2 points into +ve and -ve values



VAPNIK - CHERVONENKIS (VC) DIMENSION

Eg. 3-points Linear Classifier Model in a 2-D plane

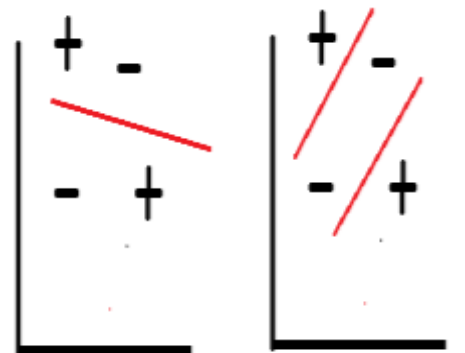
- No. of data points = $N = 3$ (+ve & -ve)
- No. of possibilities = $2^N = 2^3 = 8$
- If the points are a, b, c the 8 possibilities are $(a,b,c), (a,-b,c), (a,b,-c), (a,-b,-c), (-a,b,c), (-a,b,-c), (-a,-b,c), (-a,-b,-c)$
- A **straight line can separate 3 points** into +ve and -ve values



VAPNIK - CHERVONENKIS (VC) DIMENSION

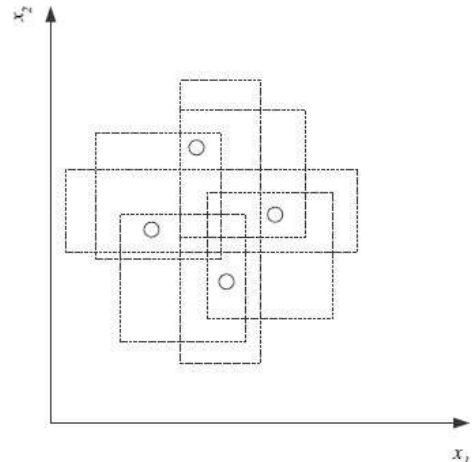
Eg. 4-points Linear Classifier Model in a 2-D plane

- No. of data points = $N = 4$
- No. of possibilities = $2^N = 2^4 = 16$
- A **straight line cannot shatter** all possibilities of 4 points into +ve and -ve values.
- **So the VC dimension of a straight line model in 2-D plane = 3.**
- $VC(H) = 3$ when H is the hypothesis class of straight lines in 2-D



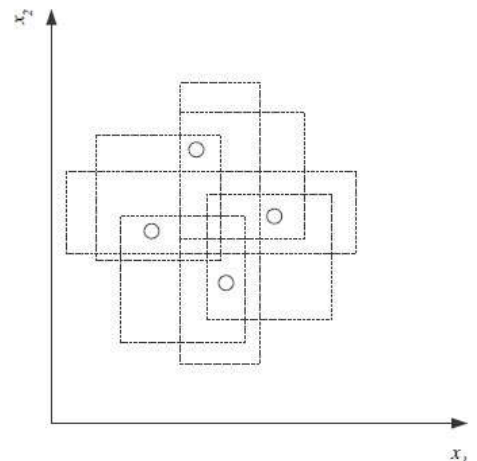
VAPNIK - CHERVONENKIS (VC) DIMENSION

- ❑ Eg. 4-points Linear Classifier Model in a 2-D plane
- ❑ An axis aligned rectangle can shatter four points in 2-D plane.
- ❑ Thus $VC(H) = 4$ when H is the hypothesis class of axis-aligned rectangles in 2-D
- ❑ Note: not necessary to be able to shatter any four points in two dimensions.
- ❑ Few rectangles covering only 2 points shown



VAPNIK - CHERVONENKIS (VC) DIMENSION

- If rectangles is the hypothesis class, only datasets containing four points can be learnt – but not so useful (because 5 points can't be shattered)
- But VC dimension is independent of the probability distribution - tells **how complex the hypothesis space is**, regardless of how the data is actually distributed.
- In real life applications, we need not worry about all possible labelings.
- So even hypothesis classes with small VC dimensions are applicable and are preferred over those with large VC dimensions (which may lead to overfitting)



PROBABLY APPROXIMATELY CORRECT (PAC) LEARNING

PROBABLY APPROXIMATELY CORRECT (PAC) LEARNING

- ❑ PAC → a framework for mathematical analysis of machine learning.
- ❑ The goal → with **high probability** (the "probably" part), the selected function will have **low generalization error** (the "approximately correct" part).
- ❑ **Definition:** Probably Approximately Correct (PAC) learning defines a mathematical relationship between the number of training samples, the error rate, and the probability that the available training data are large enough to attain the desired error rate.
- ❑ The framework helps in analyzing the **sample complexity** and **computational complexity** of learning algorithms, ensuring that they can generalize well to new, unseen data.

PROBABLY APPROXIMATELY CORRECT (PAC) LEARNING

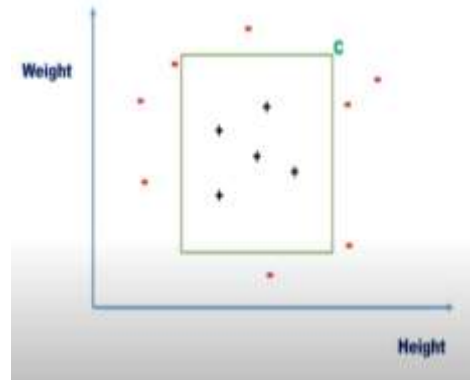
- ❑ In this framework, the learner receives N samples and must select a generalization function (called the hypothesis) from a certain class of possible functions.
- ❑ **Goal of PAC learning** → to understand the conditions under which a learning algorithm can learn a concept from a given class of concepts within a specified accuracy and confidence level.
- ❑ The learner must be able to learn the concept given any arbitrary approximation ratio, probability of success, or distribution of the samples.
- ❑ The model was later extended to treat noise (misclassified samples).

PROBABLY APPROXIMATELY CORRECT (PAC) LEARNING

- ❑ **Parameters in PAC:**
- ❑ 2 parameters used: ϵ and δ → denote error & probability respectively → must have small values
- ❑ Given a class C and examples drawn from some unknown but fixed probability distribution, $p(x)$, we want to find the number of examples, N , with probability **at least** $1 - \delta$, a system learns a concept with error **at most** ϵ where $\delta \leq \frac{1}{2}$ and $0 \leq \epsilon \leq \frac{1}{2}$
- ❑ Thus **$P(\text{Error} \leq \epsilon) \geq 1 - \delta$** (→ probability that the error is very minimum must be high)
- ❑ ϵ Gives the maximum bound on the error in accuracy with which hypotheses h is approximated → Accuracy = $1 - \epsilon$
- ❑ δ Gives the probability of failure in achieving this accuracy → confidence = $1 - \delta$.

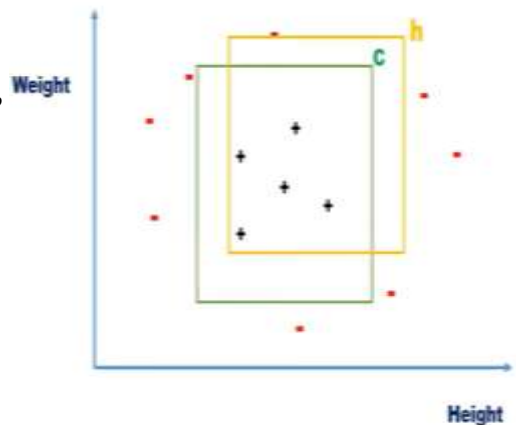
PROBABLY APPROXIMATELY CORRECT (PAC) LEARNING

- ❑ **Example:**
- ❑ To learn if a person is fit or not – based on height and weight
- ❑ The training dataset plotted as given.
- ❑ Target function $H \rightarrow$ Axis aligned rectangles



PROBABLY APPROXIMATELY CORRECT (PAC) LEARNING

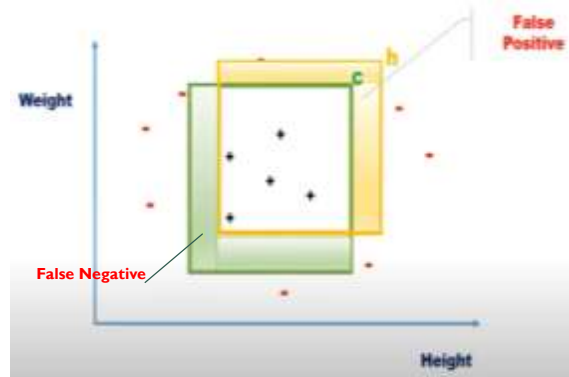
- ❑ **Example:**
- ❑ If C is the unknown class of H and h is the known one based on training set, the plot looks like this
- ❑ h closely approximates c
- ❑ Since h and c are not the same, an error region occurs



PROBABLY APPROXIMATELY CORRECT (PAC) LEARNING

Example:

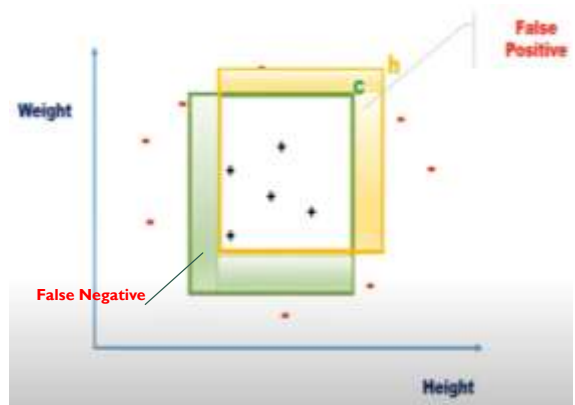
- According to h , few -ve values will be wrongly classified as +ve by $c \rightarrow$ called as False Negatives (green portion)
- Similarly According to h , few are + values, but will be wrongly classified as -ve by $c \rightarrow$ called as False Positives (yellow portion)
- Thus error occurs $\rightarrow c \text{ XOR } h$
- $P(c \text{ XOR } h) \leq \epsilon$



PROBABLY APPROXIMATELY CORRECT (PAC) LEARNING

Example:

- A H is approximately correct if $P(c \text{ XOR } h) \leq \epsilon$
- Random samples – so may be misleading – and hence there occurs a failure .
- That failure must have less probability.
- So $P(\text{Error}(h) \leq \epsilon) \geq 1 - \delta$
- $P(P(c \text{ XOR } h) \leq \epsilon) \geq 1 - \delta$



PROBABLY APPROXIMATELY CORRECT (PAC) LEARNING

Example 1:

- Consider 10 sample sets, each with different sampling distribution.
- Let $\epsilon = 0.05$ and $\delta = 0.2$
- For a hypotheses $H1$ (i.e. for a rectangle $H1$) the error($H1$) given below

Sample Set	Error ($H1$)
1	0.04
2	0.03
3	0.09
4	0.06
5	0.025
6	0.049
7	0.04
8	0.03
9	0.05
10	0.043

- Sets 1 and 2, have $\text{Error}(H1) \leq \epsilon$
- Sets 3 & 4, have $\text{Error}(H1) > \epsilon$**
- Sets 5 to 10, have $\text{Error}(H1) \leq \epsilon$
- Hence
- LHS: **$P((\text{Error}(H2))) = 8/10 = 0.8$**
- RHS: **$1 - 0.2 = 0.8$**
- $0.8 \geq 0.8$ is True**
- Hence $H1$ is Probably Approximately Correct

PROBABLY APPROXIMATELY CORRECT (PAC) LEARNING

Example 2:

- Consider 10 sample sets, each with different sampling distributions.
- Let $\epsilon = 0.05$ and $\delta = 0.2$
- For a hypotheses $H1$ (i.e. for a rectangle $H1$) the error($H1$) given below
- Check if

Sample Set	Error ($H2$)
1	0.04
2	0.035
3	0.039
4	0.06
5	0.025
6	0.059
7	0.04
8	0.03
9	0.55
10	0.043

- Sets 4, 6 & 9, have $\text{Error}(H2) > \epsilon$**
- LHS: **$P((\text{Error}(H2))) = 7/10 = 0.7$**
- RHS: **$1 - 0.2 = 0.8$**
- $0.7 \geq 0.8$ is false**
- Hence $H2$ is not Probably Approximately Correct

$$P(\text{Error}(h) \leq \epsilon) \geq 1 - \delta$$

PROBABLY APPROXIMATELY CORRECT (PAC) LEARNING

Conclusion:

- ❑ For any set of training example $T \subset X$ there may be multiple hypothesis consistent with T .
- ❑ However the learner cannot be guaranteed to choose the one corresponding to the target concept unless it is trained on every instance in X
- ❑ Learner is not required to output a zero error hypothesis but only required that error be bounded by a constant that can be made arbitrarily small

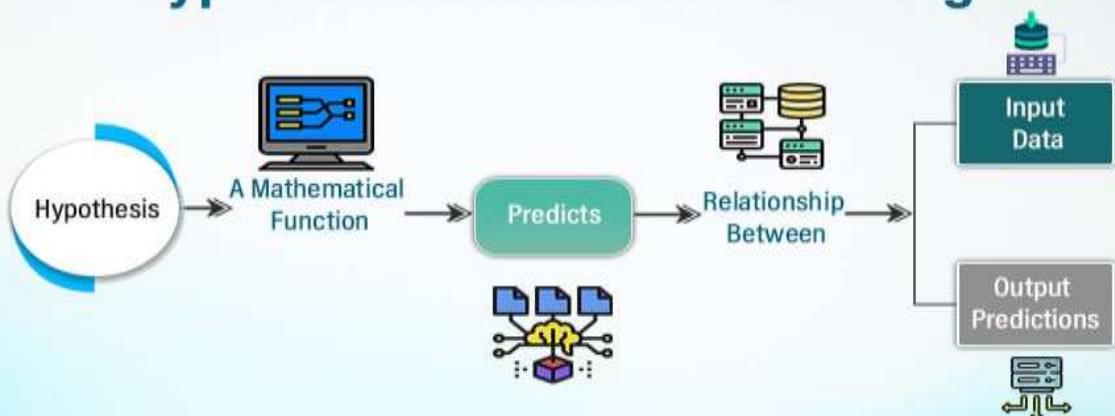
HYPOTHESIS SPACES

HYPOTHESIS SPACE

- ❑ **Hypothesis:** A hypothesis is a **function or model** that approximates the relationship between input data (features) and the target variable (output) – it maps out the inputs to the proper outputs.
- ❑ It is the candidate solution that the learning algorithm generates and refines during **training** (known inputs & known outputs – derive the mapping function) → it best maps the input to corresponding outputs with the help of an available dataset

HYPOTHESIS SPACE

Hypothesis in Machine Learning



HYPOTHESIS SPACE

- ❑ In most of the supervised learning algorithms, the goal is to find the out a possible hypothesis → it must be from the hypothesis space.
- ❑ Hypothesis space is represented by uppercase-h (H) and hypothesis by lowercase-h (h).
- ❑ **Hypothesis Space:** The set of all the possible hypothesis – from which the ML algorithm would determine the best possible function that would best describe the target function or the output(s).
- ❑ It is dependent on restrictions (criteria) and bias
- ❑ The common method to find out the possible hypothesis from the hypothesis space:

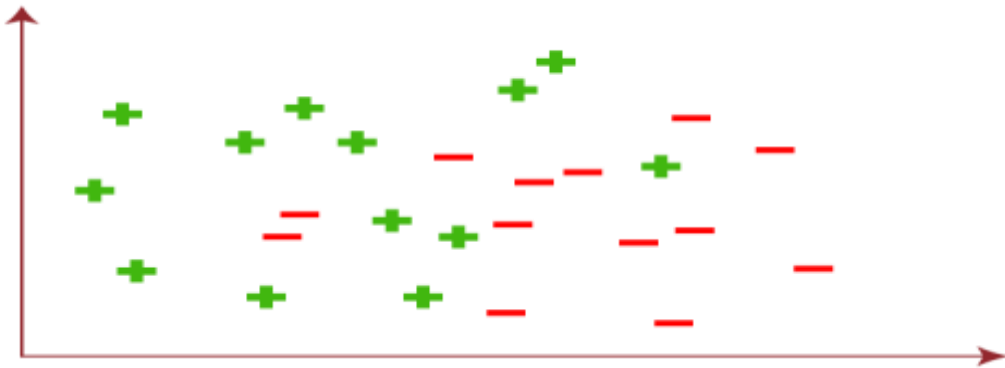
HYPOTHESIS SPACE

PRODUCING FINAL HYPOTHESIS



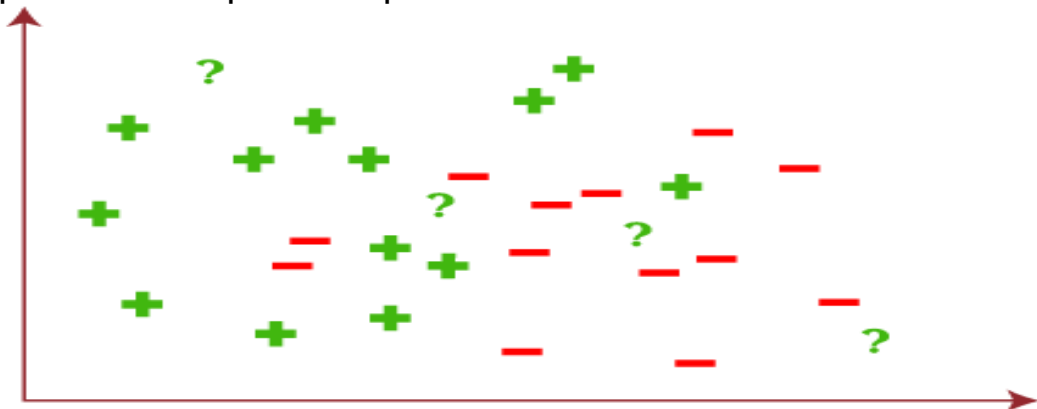
HYPOTHESIS SPACE

□ **Example:** Consider a two-dimensional coordinate plane showing the distribution of data as below:



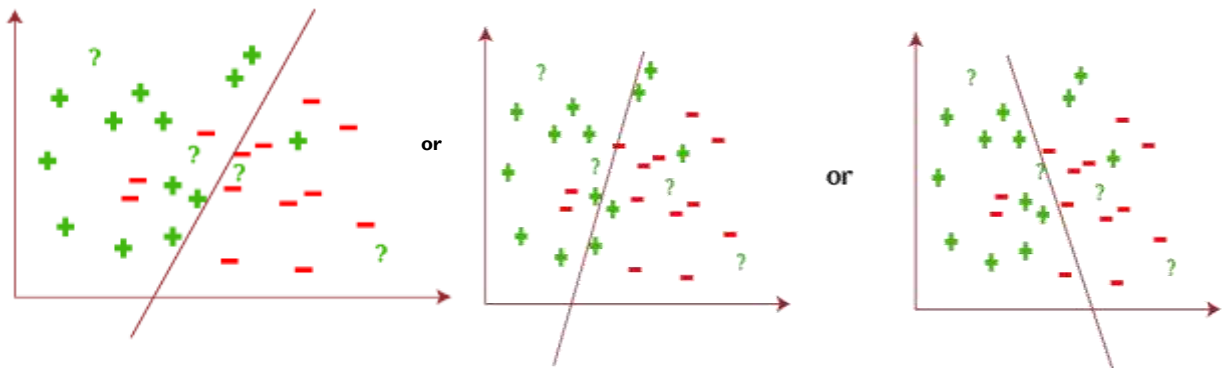
HYPOTHESIS SPACE

□ **Example:** Assume we have some test data by which ML algorithms predict the outputs for input as below:



HYPOTHESIS SPACE

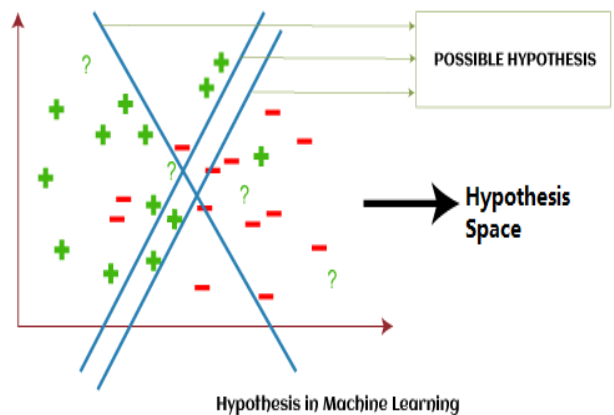
□ **Example:** this coordinate plane is divided in any of the following ways to predict output or result:



HYPOTHESIS SPACE

□ **Example:** Hypothesis space (H) is the composition of all legal best possible ways to divide the coordinate plane so that it best maps input to proper output.

□ Each individual best possible way is called a hypothesis (h). Hence, the hypothesis and hypothesis space would be like the one in the graph shown.





INDUCTIVE BIAS

INDUCTIVE BIAS

- When a cow with spots and a cowbell in Switzerland is noticed, we assume that all spotted cows in Switzerland have a cowbell.



- The picture below gives an assumption that all people in Kerala wear this costume.

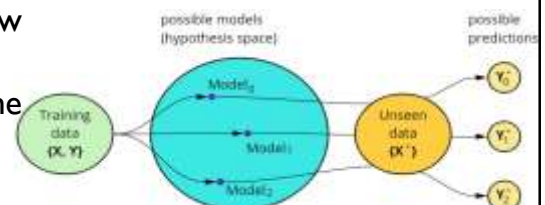


INDUCTIVE BIAS

- ❑ Data by itself is not sufficient to find the solution.
- ❑ Extra assumptions must be made to have a solution with the data we have.
- ❑ Begin with assuming a hypothesis class.
- ❑ **A learning algorithm's inductive bias** → learning bias → collection of pre-assumptions used by the learner to make learning possible and to forecast outcomes of the given new inputs that it has never seen before.
- ❑ **Inductive bias** → refers to the set of assumptions a learning algorithm makes to generalize beyond the training data.
- ❑ In traditional machine learning, every algorithm has its own inductive biases.

INDUCTIVE BIAS

- ❑ Inductive bias helps guide the algorithm in choosing **one hypothesis over others** that may also fit the training data.
- ❑ Model selection is about choosing the right inductive bias.
- ❑ Preferring one solution over another after viewing certain instances sums up inductive bias.
- ❑ Different models can be trained based on fixed training data.
- ❑ All those models will behave differently for new unseen data (future data).
- ❑ Inductive bias makes the algorithm learn one pattern instead of another pattern.
- ❑ Inductive bias **enables generalization**.



INDUCTIVE BIAS - EXAMPLES OF INDUCTIVE BIAS

- ❑ In linear regression → assumption that the variable (Y) is linearly dependent on the explanatory variables (X)
- ❑ In KNN → assumption that similar data points are clustered near each other away from the dissimilar ones.
- ❑ In Bayesian Modeling → the priors selected reveal the bias about what will happen for unseen data when much data is not available
- ❑ Learning the class of family car → many ways of separating the positive examples from the negative examples → assuming the shape of a rectangle is an inductive bias

INDUCTIVE BIAS - ADVANTAGES OF A SIMPLE MODEL

- ❑ A complex model may not be making any errors in prediction
- ❑ But using a simple model, has certain advantages:
 - A simple model is
 - Easy to use.
 - Easy to train. (Because of fewer parameters possible, it is easier to find the corner values of a rectangle than the control points of an arbitrary shape).
 - Easy to explain.
 - Generalize better than a complex model → This principle is known as Occam's razor.

INDUCTIVE BIAS – EXAMPLE – OCCAM'S RAZOR

- ❑ Problem-solving principle that recommends searching for explanations constructed with the smallest possible set of elements.
- ❑ States that simpler explanations are more reasonable and any unnecessary complexity should be shaved off.
- ❑ Assumption → the simplest consistent hypothesis about the target function is actually the best. (Consistent → means that the hypothesis of the learner yields correct outputs for all of the examples that have been given to the algorithm)
- ❑ Given 2 models with either of the following characteristics, prefer the simpler one.
 - same generalization error or
 - same training-set error,
 - same performance on the testing dataset

INDUCTIVE BIAS

TYPE	DESCRIPTION
Maximum conditional independence	if the hypothesis can be cast in a Bayesian framework, try to maximize conditional independence
Minimum cross-validation error	when trying to choose among hypotheses, select the hypothesis with the lowest cross-validation error. Although cross-validation may seem to be free of bias
Maximum margin	when drawing a boundary between two classes, attempt to maximize the width of the boundary.
Minimum description length	when forming a hypothesis, attempt to minimize the length of the description of the hypothesis.
Minimum features	unless there is good evidence that a feature is useful, it should be deleted.
Nearest neighbors	assume that most of the cases in a small neighborhood in feature space belong to the same class.

GENERALIZATION

GENERALIZATION

- ❑ **Generalization** → The ability of a ML model to perform well on new, unseen data, after being trained on a limited set of training examples.
- ❑ How well the concepts learned by a machine learning model apply to specific unseen examples.
- ❑ Goal of a good ML model → A ML model generalizes well if it can apply what it learned during training to correctly predict or classify data it has never seen before, in real-world and unpredictable scenarios
- ❑ Learn patterns that **apply broadly**.
- ❑ The model having the **best generalization** should be selected.
- ❑ Eg. for generalization → A spam email classifier → generalization enables the model to identify the underlying patterns and characteristics that distinguish spam from legitimate emails. (Without generalization, the model may become too specific to the training set, memorizing specific words or phrases that were common in the training data and failing to understand new examples)

GENERALIZATION

❑ Causes for poor performance of ML algorithms:

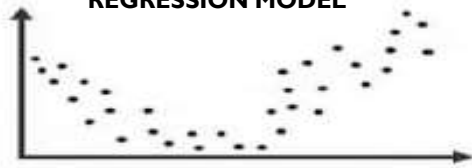
- **Underfitting** → The production of a ML model that is not complex enough to accurately capture relationships between a dataset's entire set of features and a target variable – not trained enough → underfit models exhibit high error on both the training and testing data and also a high bias.
- **Overfitting** → The production of an analysis which corresponds **too closely or exactly to a particular set of data**, and may therefore fail to fit additional data or predict future observations reliably → perform exceptionally well on training data but fail to generalize to new, unseen data → overly sensitive model → poor generalization causes overfitting

GENERALIZATION – ISSUES / CHALLENGES

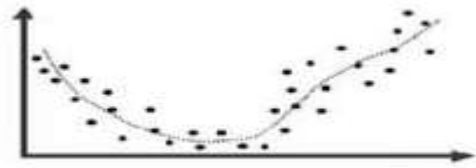
Sl. No.	Issues / Challenges	Meaning
1.	Training error	Error on data the model was trained on
2.	Test error	Error on new/unseen data
3.	Overfitting	Low training error but high test error → poor generalization
4.	Underfitting	High error on both training and test data → model too simple
5.	Bias-Variance Tradeoff	Finding the balance to ensure good generalization

GENERALIZATION - EXAMPLE

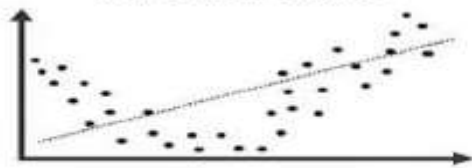
REGRESSION MODEL



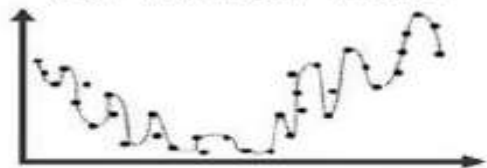
(a) Given dataset



(b) "Just right" model



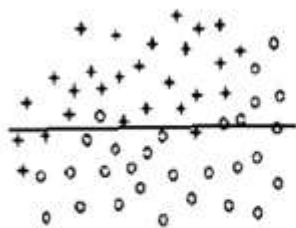
(c) Underfitting model



(d) Overfitting model

GENERALIZATION - EXAMPLE

CLASSIFICATION MODEL



(a) Underfitting



(b) Right fitting



(c) Overfitting

GENERALIZATION – CROSS VALIDATION

- ❑ It is needed to measure the generalization ability of a hypothesis - the quality of its inductive bias
- ❑ Procedure used → cross-validation.
 - ❑ Simulate this by dividing the large enough training set into two parts.
 - ❑ Use one part for training (to find a hypothesis / model)
 - ❑ Use the remaining part to test the generalization ability → the validation set.
 - ❑ The hypothesis that is the most accurate on the validation set is the best one (the one that has the best inductive bias).

GENERALIZATION – CROSS VALIDATION

Techniques used to Improve Generalization:

- ❑ Cross-validation
- ❑ Regularization (L1, L2)
- ❑ Early stopping
- ❑ Pruning (in decision trees)
- ❑ Using more training data
- ❑ Reducing model complexity



BIAS VARIANCE TRADE-OFF



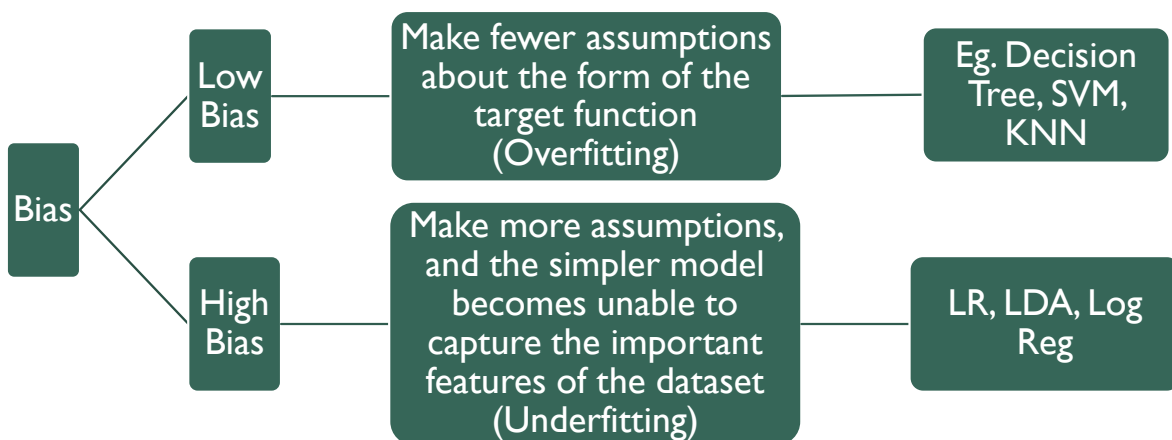
BIAS AND VARIANCE

- ❑ Two key sources of error that affect how well a model performs, especially on unseen data.
- ❑ Help us understand and manage underfitting and overfitting.
- ❑ Bias → errors due to incorrect assumptions in the learning algorithm.
- ❑ Variance → errors due to model sensitivity to small changes in training data.

BIAS

- ❑ A ML model analyses the data, find patterns in it and make predictions.
- ❑ Bias → The difference between the prediction of the values by the ML model and the actual / expected values.
- ❑ Tells that how much a **predicted value** is different from its actual value
- ❑ An inability of ML algorithms to capture the true relationship between the data points.
- ❑ Each algorithm begins with some amount of bias because bias occurs from assumptions in the model, which makes the target function simple to learn.
- ❑ **High bias** → More differences → causes error in training as well as testing data → results in **underfitting** model (because of too simple hypothesis / model)
- ❑ **Low bias** → results in **overfitting** model

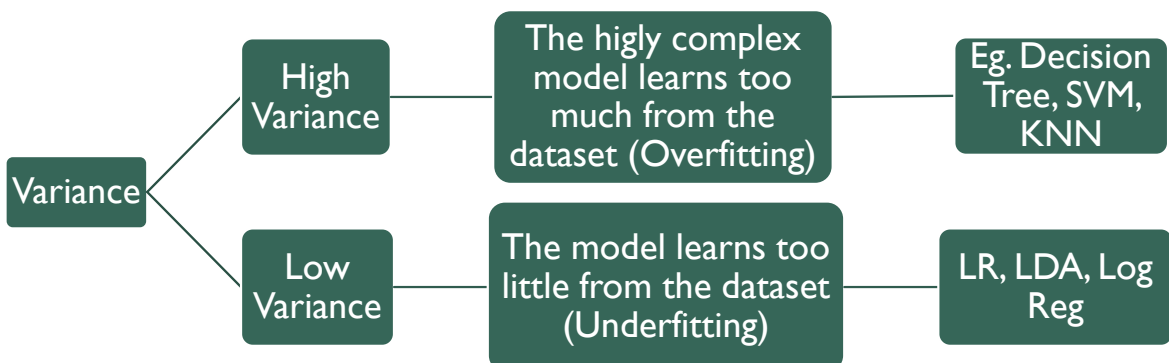
BIAS



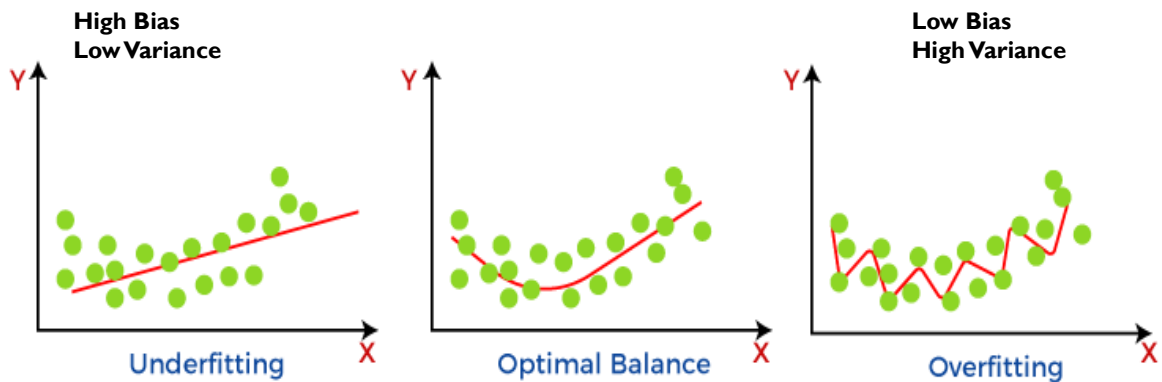
VARIANCE

- ❑ **Variance** → The variability of model prediction for a given data point which tells us the spread of our data.
- ❑ Specifies the amount of variation in the prediction if the different training data was used.
- ❑ Tells that how much a **random variable** is different from its expected value
- ❑ **Low variance** → there is a small variation in the prediction of the target function with changes in the training data set → results in **underfitting** model → High training error and similar test error
- ❑ **High variance** → shows a large variation in the prediction of the target function with changes in the training dataset → results in **overfitting** model (fitting the training set accurately for the training data and high order hypothesis but is not the solution as the error with unseen data is high) → Low training error and high test error → too complex model

VARIANCE



BIAS & VARIANCE

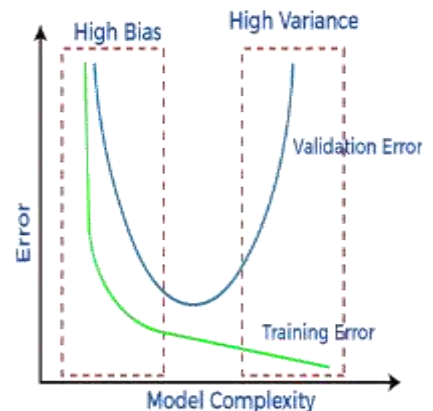


BIAS & VARIANCE

- ❑ Bias & variance → 2 sources of error in a predictive model.
- ❑ The bias error → an error from erroneous assumptions in the learning algorithm.
- ❑ The variance → an error from sensitivity to small fluctuations in the training set.
- ❑ High bias causes underfitting model
- ❑ High variance causes overfitting model
- ❑ Take care of bias and variance in order to avoid overfitting and underfitting in the model.

BIAS – VARIANCE TRADE-OFF

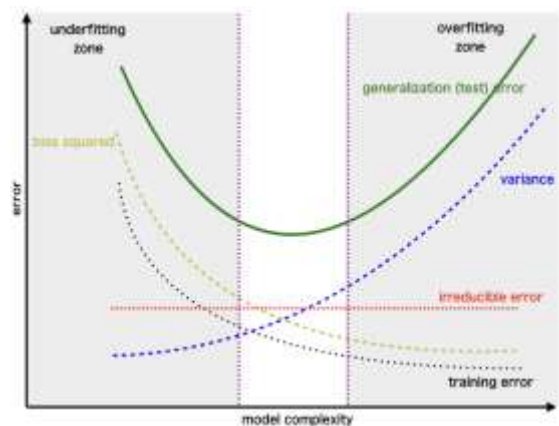
- ❑ Simpler model with fewer parameters → may have high bias and low variance → error prone.
- ❑ Complex model with large number of parameters (hence a hypothesis with high degree equation) → will have low bias and high variance → not good for test / new data
- ❑ So make a balance between bias and variance errors
- ❑ **Bias-Variance trade-off** → describes the relationship between a model's complexity, the accuracy of its predictions, and how well it can make predictions on previously unseen data that were not used to train the model.



BIAS – VARIANCE TRADE-OFF

- ❑ Model with low variance and low bias → impractical.
- ❑ Model with high variance and high bias → predictions are inconsistent and also inaccurate on average.
- ❑ Lower variance causes higher bias
- ❑ Lower bias causes higher variance
- ❑ The best fit will be given by the hypothesis on the tradeoff point.
- ❑ The goal → to find a balance — not too simple, not too complex

BIAS AND VARIANCE AS FUNCTION OF MODEL COMPLEXITY



BIAS – VARIANCE TRADE-OFF

BIAS	VARIANCE	PARAM ETERS	ASSUM PTIONS	TRAININ G ERROR	TESTING ERROR	FITTING MODEL	MODEL COMPL EXITY
Low	High	Many	Less	Less	More	Overfitting	Complex Model
High	Low	Few	More	More	More	Underfitting	Simple Model